

Measuring “Dark Matter” in Asset Pricing Models

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Outline

1. Motivation: A Question About Model Fragility
2. Motivating Example: Gordon Growth
3. General Framework: One Measure and Three Theorems
4. Quantitative Examples: Three Asset-Pricing Models
 - 4.1 Rare Disaster Risk
 - 4.2 Time-Varying Disaster Risk
 - 4.3 Long-Run Risk
5. Conclusion: Diagnose Before You Estimate

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From Cosmology to Asset Pricing: An Analogy

- ▶ In cosmology, dark matter is invisible, yet required to make Einstein's general relativity consistent with the observed motion of stars.
- ▶ Campbell (2008): rare disasters are the “dark matter for economists.”
- ▶ A working definition of *economic* dark matter: a model component that
 - ▶ cannot be directly measured from fundamental data (consumption, output);
 - ▶ is identified only via cross-equation restrictions.
- ▶ Question: can we quantify how much dark matter a model relies on?

Two Contributions: A Measure and Three Theorems

- ▶ A scalar dark-matter measure $\varrho(\theta_0)$, computable for any GMM-based structural model.
- ▶ Three theorems linking $\varrho(\theta_0)$ to model fragility:
 - ▶ Higher $\varrho(\theta_0) \Rightarrow$ lower internal refutability (specification tests lose power).
 - ▶ Higher $\varrho(\theta_0) \Rightarrow$ higher in-sample overfitting.
 - ▶ Recursive GMM is robust regardless of $\varrho(\theta_0)$.

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Setup: A Lucas Tree with Two GMM Moments

- ▶ Dividend dynamics:

- ▶ $\frac{Y_t}{Y_{t-1}} = 1 + \theta + \sigma_Y \varepsilon_{Y,t}$, with $\mathbb{E}[\varepsilon_{Y,t}] = 0$, $\mathbb{E}[\varepsilon_{Y,t}^2] = 1$.

- ▶ Price-dividend ratio (Gordon growth):

- ▶ $\frac{P_t}{Y_t} = F(\theta) + \sigma_P \varepsilon_{P,t}$, where $F(\theta) = \frac{1 + \theta}{r - \theta}$.

- ▶ Two GMM moments:

$$\mathbb{E} \begin{bmatrix} \sigma_Y^{-1} (Y_t/Y_{t-1} - 1 - \theta) \\ \sigma_P^{-1} (P_t/Y_t - F(\theta)) \end{bmatrix} = 0. \quad (1)$$

- ▶ Top row $m^{(1)}$: **baseline** (fundamentals only).
- ▶ Bottom row $m^{(2)}$: **cross-equation restriction**.
- ▶ Together $m = (m^{(1)}, m^{(2)})^\top$: the **full model**.

C-Test: Comparing Baseline and Full Fit

- ▶ Local misspecification (Pitman drift):

$$\mathbb{E}\left[m^{(2)}(\theta_0)\right] = \frac{\lambda}{\sigma_P\sqrt{n}}. \quad (2)$$

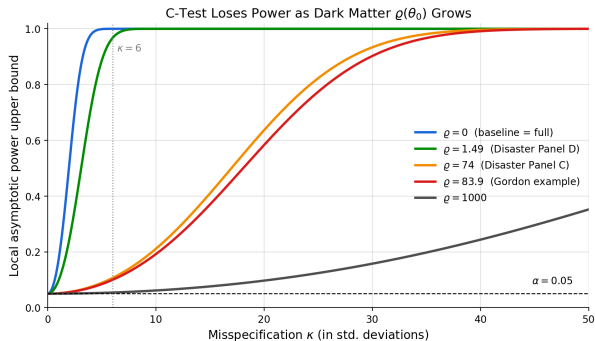
- ▶ The drift shrinks at rate $1/\sqrt{n}$, so multiplied by the GMM scale $\sqrt{n}$ the bias remains finite ($= \lambda/\sigma_P$).
- ▶ C-test (Eichenbaum, Hansen, Singleton 1988):

$$C_n \equiv \min_{\theta} n |\hat{m}_n(\theta)|^2 - \min_{\theta} n |\hat{m}_n^{(1)}(\theta)|^2. \quad (3)$$

- ▶ Asymptotic distribution:

$$C_n \xrightarrow{d} \left[Z + \frac{\lambda}{\sigma_P\sqrt{1 + \varrho(\theta_0)}} \right]^2, \quad Z \sim \mathcal{N}(0, 1). \quad (4)$$

C-Test Loses Power as $\varrho(\theta_0)$ Grows



- ▶ For $\varrho(\theta_0) = 83.9$: a $\kappa = 6$ standard-deviation misspecification is detected with power at most 0.089.
- ▶ As $\varrho(\theta_0) \rightarrow \infty$, power $\rightarrow \alpha$: the test becomes useless.

High dark matter $\Rightarrow$ low refutability, regardless of sample size.

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Splitting Parameters: Shared and Nuisance

- ▶ Recursive moment structure splits parameters into two blocks:

$$m(y_{t-1}, y_t, \theta) = \begin{pmatrix} m^{(1)}(\cdot, \theta^{(1)}) \\ m^{(2)}(\cdot, \theta^{(1)}, \theta^{(2)}) \end{pmatrix}, \quad \theta = \begin{pmatrix} \theta^{(1)} \\ \theta^{(2)} \end{pmatrix}. \quad (5)$$

- ▶ $\theta^{(1)}$: **shared** parameters (identified by both baseline and full).
 - ▶ $\theta^{(2)}$: **nuisance** parameters (identified only via cross-equation).
- ▶ Block-triangular Jacobian:

$$D(\theta) = \begin{pmatrix} D_{11} & 0 \\ D_{21} & D_{22} \end{pmatrix}, \quad D_{ij} \equiv \mathbb{E} \left[\partial m^{(i)} / \partial \theta^{(j)} \right] \big|_{\theta_0}. \quad (6)$$

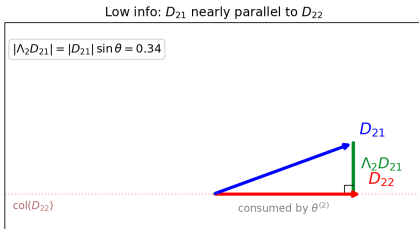
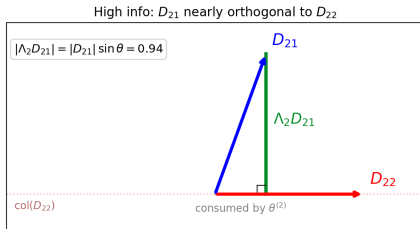
- ▶ Convention: $m^{(1)}$ summarizes consumption / output dynamics; $m^{(2)}$ adds pricing equations.

Λ_2 : The Geometry of Information Decomposition

- Information matrix decomposes into baseline plus a projected term:

$$\mathcal{I}_F = \underbrace{D_{11}^\top D_{11}}_{=\mathcal{I}_B} + D_{21}^\top \Lambda_2 D_{21}, \quad \Lambda_2 = I - D_{22}(D_{22}^\top D_{22})^{-1} D_{22}^\top. \quad (7)$$

Λ_2 projects D_{21} onto the orthogonal complement of D_{22}



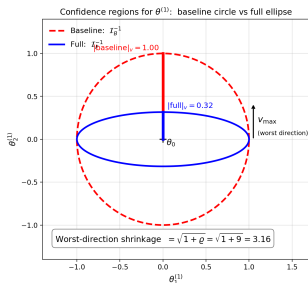
- $D_{21}^\top \Lambda_2 D_{21}$: information about $\theta^{(1)}$ that survives after the nuisance is consumed.

Dark Matter as a Worst-Direction Variance Ratio

- Definition (paper Definition 1):

$$\varrho(\theta_0) \equiv \max_{|v|=1} \frac{v^\top \mathcal{I}_B^{-1} v}{v^\top \mathcal{I}_F^{-1} v} - 1. \quad (8)$$

- Numerator $v^\top \mathcal{I}_B^{-1} v$: asymptotic variance of the efficient **baseline** GMM estimator of $v^\top \theta^{(1)}$.
- Denominator $v^\top \mathcal{I}_F^{-1} v$: same quantity for the **full** estimator.
- Ratio measures how much the cross-equation restriction shrinks the variance of $v^\top \theta^{(1)}$.



Three Theorems: Dark Matter Drives Fragility

- ▶ **Theorem 1 (Refutability)**: maximin local power of any size- α GMM specification test is bounded above by a function decreasing in $\varrho(\theta_0)$ and approaches α as $\varrho(\theta_0) \rightarrow \infty$.

- ▶ **Theorem 2 (Overfitting of efficient GMM)**:

$$\text{worst-case overfitting} = d_{\theta,1} + c(\pi) \varrho(\theta_0) \kappa^2. \quad (9)$$

- ▶ **Theorem 3 (Recursive GMM is robust)**:

$$\text{worst-case overfitting} = d_{\theta,1}, \text{ independent of } \varrho(\theta_0). \quad (10)$$

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Rare Disaster: Setup and Parameters of Interest

- ▶ State variable $z_t \sim \text{Bernoulli}(p)$ (Rietz 1988, Barro 2006).
- ▶ Normal regime ($z_t = 0$): consumption growth $g_t = u_t \sim \mathcal{N}(\mu, \sigma^2)$.
- ▶ Disaster regime ($z_t = 1$): consumption growth $g_t = -v_t$, where

$$v_t \sim \mathbf{1}\{v_t > \underline{v}\} \xi e^{-\xi(v_t - \underline{v})}, \quad \underline{v} > 0, \xi > 0. \quad (11)$$

- ▶ Excess log return r_t :
 - ▶ Normal regime: $r_t = \eta + \rho(\tau/\sigma)(g_t - \mu) + \sqrt{1 - \rho^2} \tau \varepsilon_{0,t}$.
 - ▶ Disaster regime: $r_t = \ell g_t + \varsigma \varepsilon_{1,t}$.
- ▶ Representative agent has CRRA utility, $u(c) = c^{1-\gamma}/(1-\gamma)$.
- ▶ Parameters of interest:
 - ▶ $\theta^{(1)} = (p, \xi)$: probability and inverse mean disaster size.
 - ▶ $\theta^{(2)} = \emptyset$: no nuisance in this example.

Moment Conditions and Auxiliary Parameters

- ▶ GMM moment vector for $\theta = (p, \xi)^\top$:

$$m_t(\theta) = \Omega(\theta)^{-1/2} \begin{pmatrix} z_t - p \\ g_t - (1 - z_t)\mu + z_t(\underline{v} + 1/\xi) \\ r_t - (1 - z_t)[\eta + \rho(\tau/\sigma)(g_t - \mu)] - z_t \ell g_t \end{pmatrix}. \quad (12)$$

- ▶ First two rows: **baseline** (consumption regime + disaster frequency).
 - ▶ Third row: **cross-equation restriction** (asset pricing).
-
- ▶ Auxiliary parameters fixed at empirical values (NIPA + CRSP, 1929–2011):
 - ▶ $(\mu, \sigma, \tau, \varsigma, \rho, \underline{v}, \ell, \gamma) = (1.87\%, 1.95\%, 19.14\%, 34.89\%, 0.59, 7\%, 3, [3, 10, 24])$.

Closed-Form Dark Matter for Rare Disaster

- ▶ Closed-form equity premium:

$$\bar{r}(p, \xi) = (1 - p) \eta - p \ell (\underline{v} + 1/\xi), \quad (13)$$

where η involves $\Delta(\xi) = \xi \left[e^{\gamma \underline{v}} / (\xi - \gamma) - e^{s^2/2 + (\gamma - \ell) \underline{v}} / (\xi + \ell - \gamma) \right]$.

- ▶ Closed-form dark matter:

$$\varrho(\theta_0) = 1 + \frac{p \Delta(\xi)^2 + p(1 - p) \xi^2 \dot{\Delta}(\xi)^2}{(1 - \rho^2) \tau^2 (1 - p)^2} e^{2\gamma \mu - \gamma^2 \sigma^2}. \quad (14)$$

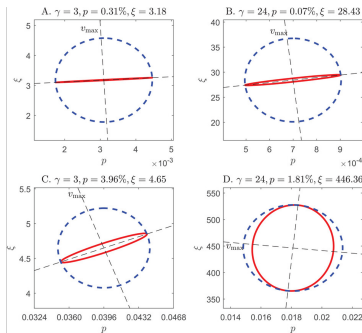
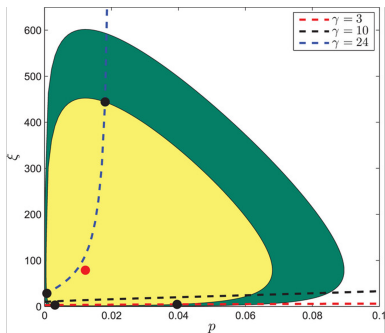
- ▶ Key driver:

- ▶ $\Delta(\xi)$ explodes as $\xi \rightarrow \gamma^+$ because of the $1/(\xi - \gamma)$ pole.
- ▶ Its derivative $\dot{\Delta}(\xi)$ has a $1/(\xi - \gamma)^2$ pole, even faster.
- ▶ Result: **rare-and-large disasters (small p , small ξ) explode dark matter.**

Same Equity Premium, Vastly Different Fragility

- All four calibrations match the same equity premium $\bar{r} = 5.09\%$.

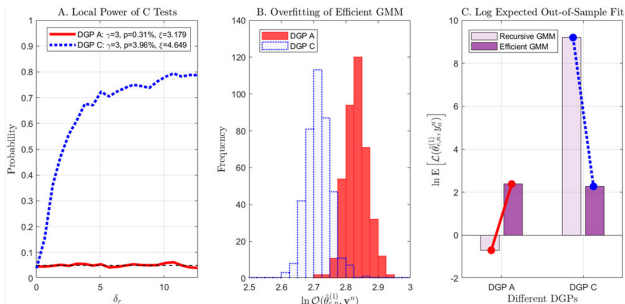
Cal.	Description	p	ξ	$\varrho(\theta_0)$
A	Rare & large, $\gamma = 3$	0.31%	3.18	1.78×10^4
B	Rare & mid, $\gamma = 24$	0.07%	28.4	5.6×10^2
C	Frequent & large, $\gamma = 3$	3.96%	4.65	74
D	Frequent & small, $\gamma = 24$	1.81%	446	1.49



Monte Carlo Confirms All Three Theorems

- ▶ Local DGP with mid-sample structural break:

$$\bar{r}_n = \bar{r}(p_0, \xi_0) + \frac{\iota_t \delta_r}{\sqrt{n}}, \quad \iota_t = +1 \text{ for } t \leq \pi n, \text{ else } -1. \quad (15)$$



- ▶ Panel A $\Rightarrow$ **Theorem 1**: power flat for high $\varrho(\theta_0)$, steep for low $\varrho(\theta_0)$.
- ▶ Panel B $\Rightarrow$ **Theorem 2**: efficient-GMM overfit much larger for high $\varrho(\theta_0)$.
- ▶ Panel C $\Rightarrow$ **Theorem 3**: recursive GMM wins under high $\varrho(\theta_0)$.

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Time-Varying Disaster: Setup and Parameters

- ▶ Wachter (2013): Epstein–Zin preferences with EIS = 1:

$$\ln V_t = (1 - \delta) \ln C_t + \delta(1 - \gamma)^{-1} \ln \mathbb{E}_t \left[V_{t+1}^{1-\gamma} \right]. \quad (16)$$

- ▶ Consumption with time-varying disaster:

- ▶ $\Delta c_{t+1} = \mu + \sigma_c \varepsilon_{c,t+1} - \zeta_{t+1}$, where $\zeta_{t+1} = z_{t+1} v_{t+1}$.
- ▶ v_{t+1} truncated exponential with rate ξ , lower bound $\underline{v}$.
- ▶ $z_{t+1} \sim \text{Bernoulli}(p_t)$ with $p_t = \max(\underline{p}, \tilde{p}_t)$.

- ▶ Disaster probability follows AR(1):

$$\tilde{p}_{t+1} = (1 - \rho) \bar{p} + \rho \tilde{p}_t + \sigma_p \sqrt{\bar{p}_t} \varepsilon_{p,t+1}. \quad (17)$$

- ▶ Levered dividend

$$\Delta d_{t+1} = \mu - \frac{1}{2} \varphi^2 \sigma_c^2 + \phi \sigma_c \varepsilon_{c,t+1} - \phi \zeta_{t+1} + \varphi \sigma_c \varepsilon_{d,t+1}. \quad (18)$$

- ▶ Parameters:

- ▶ $\theta^{(1)} = (\mu, \sigma_c^2, \bar{p}, \rho, \sigma_p^2, \xi, \phi, \varphi, q)$ (9-dim).
- ▶ $\theta^{(2)} = \gamma$ (risk aversion, treated as nuisance).

Two Calibrations and the Worst-Direction (TVD)

- Two calibrations match every empirical moment in Panel B equally well:

	$\varrho(\theta_0)$	univariate $\varrho(\theta_0)$ for individual parameters of $\theta^{(1)}$							
		μ	σ_c^2	$\bar{p}$	ρ	σ_p^2	ξ	ϕ	φ
M1 (Wachter 2013)	1.49×10^3	1.99	1.98	1.48	262	165	2.10	2.04	1.97
M2 (alternative)	80.99	2.35	2.33	1.98	1.31	1.26	2.88	2.43	2.33

- Worst-case 1D $\varrho(\theta_0)$ is far larger than any univariate value.
- ρ : the persistence of time-varying disaster probability.
- σ_p^2 : the variance of time-varying disaster probability shocks.
- M1 has $\rho = 0.936$ and $\sigma_p = 0.067$.
- M2 has $\rho = 0.0394$ and $\sigma_p = 0.85$.
 - Lower persistence + higher conditional vol $\Rightarrow$ 19 $\times$ smaller dark matter, same data fit.

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Long-Run Risk: Setup and Parameters

- ▶ Bansal, Kiku, Yaron (2012): full Epstein–Zin preferences,

$$V_t = \left[(1 - \delta) C_t^{1-1/\psi} + \delta (\mathbb{E}_t V_{t+1}^{1-\gamma})^{(1-1/\psi)/(1-\gamma)} \right]^{1/(1-1/\psi)}. \quad (19)$$

- ▶ Consumption dynamics:

- ▶ $\Delta c_{t+1} = \mu_c + x_t + \sigma_t \epsilon_{c,t+1}$.
- ▶ $x_{t+1} = \rho x_t + \varphi_x \sigma_t \epsilon_{x,t+1}$ (long-run growth, persistence ρ).
- ▶ $\sigma_{t+1}^2 = \bar{\sigma}^2 + \nu(\sigma_t^2 - \bar{\sigma}^2) + \sigma_w \epsilon_{\sigma,t+1}$ (stochastic volatility, persistence ν).

- ▶ Dividend:

$$\Delta d_{t+1} = \mu_d + \phi_d x_t + \varphi_{d,c} \sigma_t \epsilon_{c,t+1} + \varphi_{d,d} \sigma_t \epsilon_{d,t+1}. \quad (20)$$

- ▶ Parameters:

- ▶ $\theta^{(1)} = (\mu_c, \rho, \varphi_x, \bar{\sigma}^2, \nu, \sigma_w, \mu_d, \phi_d, \varphi_{d,c}, \varphi_{d,d})$ (10-dim).
- ▶ $\theta^{(2)} = (\gamma, \psi)$.

Two Calibrations and the Worst-Direction (LRR)

- ▶ Two calibrations match every empirical moment equally well:

	$\varrho(\theta_0)$	univariate $\varrho(\theta_0)$ for individual parameters of $\theta^{(1)}$						
		μ_c	ρ	φ_x	$\bar{\sigma}^2$	ν	σ_w	ϕ_d
M1 (BKY 2012)	196.3	1.0	1.1	1.0	49	98	1.0	3.4
M2 (alternative)	21.1	1.0	1.1	1.0	1.0	3.4	1.0	4.2

- ▶ ν : the persistence of conditional variance of consumption growth.
- ▶ $\bar{\sigma}^2$: the long-run variance of consumption growth.
- ▶ M1 has $\nu = 0.999$ (near unit root) and $\gamma = 10$.
- ▶ M2 has $\nu = 0.98$ and $\gamma = 27$.
 - ▶ Stepping back from the unit root $\Rightarrow$ 9 $\times$ smaller dark matter.

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Take-Aways: A Thermometer for Model Fragility

- ▶ Dark matter is a **model property**, not a small-sample issue: it survives $n \rightarrow \infty$.
- ▶ High $\varrho(\theta_0)$ implies low refutability and high overfitting (Theorems 1 and 2).
- ▶ Practical guidance:
 - ▶ Pick robust calibrations (small $\varrho(\theta_0)$) when several fit equally well.
 - ▶ Estimate with **recursive GMM** when $\varrho(\theta_0)$ is large (Theorem 3).
- ▶ $\varrho(\theta_0)$ is comparable within a model family, not across functional forms.

Before celebrating that a model fits the data, ask: **how much dark matter is doing the work?**

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